Tatiana Chibisova

Graphics & 3D Vision Engineer
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EDUCATION

09.2021 ~ 02.2024 **KAIST** ***M.S.*** *Daejeon, South Korea*
Supervisor: Sung Hee Lee • Graduate School of Culture Technology
Honors: KAIST Scholarship

09.2017 ~ 06.2021 **HSE University** ***B.S.*** *Moscow, Russia*
Supervisor: Nadejda Trubochkina • Computer Engineering, Intelligent Robotics
Honors: Academic Tuition Waiver due to Academic Excellence

PERSONAL PROJECTS

Avatar Playground
https://github.com/chibisova/AvatarPlayground

- Built a modular C++17/OpenCV monocular visual odometry and 3D reconstruction pipeline (feature matching, essential matrix + RANSAC, PnP localization, sparse mapping, camera trajectory estimation); validated on TUM RGB-D benchmark, 84.3% frame-pair acceptance rate, 2.02° mean rotation error.
- Currently extending toward parametric human mesh recovery (MHR, GNM).

MHR Mesh Viewer
C++, Qt, OpenGL, Python · repository coming soon

- Built an interactive Qt/OpenGL viewer for the MHR parametric human model with PCA shape and pose sliders and real-time mesh deformation.
- Developing real-time measurement-driven fitting in Python: adjusts MHR shape parameters to match target bust, waist and hip values via live Jacobian updates.

WORK EXPERIENCE

03.2024 ~ 01.2026 **Skonec Entertainment** *Seoul, South Korea*
Technical Artist

- Shader Development: developed interactive real-time custom shaders
- Optimization: reduced draw calls **4x** per scene, achieving stable **70 FPS**
- UX/UI: designed avatar customization interface and MR snapshot feature
- 3D/2D Assets Production: cut game-ready art assets production time from weeks-months to **1 week** by utilizing AI tools

09.2021 ~ 02.2024 **KAIST (LAVA Lab)** *Daejeon, South Korea*
Graduate Research Student

- Set up and evaluated **FLAME**-based audio-driven facial animation, identifying failure modes in emotional expressiveness
- Proposed **audio-to-motion** reconstruction (piano performance → virtual pianist motion) as a research direction; synchronized audio and mocap data by onset-peak alignment to build the audio-to-motion visualization

09.2019 ~ 12.2019 **CROC** *Moscow, Russia*
Junior Java Software Developer

- Built real-time data collection and Java processing logic for medical hardware
- Built an interactive web interface (Vue.js) for hardware data visualization

SKILLS

Graphics & XR: Unity, Shader Graph/HLSL, VR/AR Prototyping, Real-Time VFX, C#, OpenGL
Research & Code: Python / PyTorch, C++, OpenCV, Signal Processing
3D & Art: Blender, ZBrush, Substance Painter, Photoshop / Illustrator, Figma
Research Interests: 3D Vision, Parametric Human Models

LANGUAGES

English: Business proficient (TOEIC 955)
Korean: Conversational (TOPIK 3 급)
Russian: Native

AWARDS

2023 **Korea Metaverse Developer Contest (Ministry of Science and ICT)**
Grand Prize, Space Cleaners VR

2023 **Daejeon Startup Program**
Excellence Award, NoraeVerse: AI vocal-coaching VR karaoke platform

2022 **ACM CHI, Student Game Competition**
Finalist, "Play With Your Emotions" (published in CHI '22 Extended Abstracts)

2019 **InnovateFPGA, Terasic / Intel**
Regional Finalist, DE10-Nano-based smart home system

2018 **Avito & Android Academy Hackathon**
Excellence Award (User Care track)

ACTIVITIES

2026 **Zerobase Korea**
Founder, workflow tool for cross-discipline (art/dev) game production pipelines

2025 **GameAiCON Seoul**
Invited Speaker, "The Impact of Generative AI on the Game Ecosystem"

2025 **Evacuation Order (疊開), Jeju 4.3, dir. Lim Heung-soon**
Technical Artist, interactive 3D installation

2024 **The Memorial Shower, Jeju 4.3, ARKO Public Art Project**
Technical Artist, game adaptation of Lim Heung-soon's film

2022 **SIGGRAPH Asia**
Student Volunteer, Daegu